ORBITAL MECHANICS OF PLANETS — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 29, 2026

Diagram-based questions on Orbital Mechanics of Planets

Question 1

10.0 points

Examine the diagram below depicting a planet's elliptical orbit around the sun. Using Kepler's second law, describe how the area swept out by a line from the planet to the sun varies over time. Explain how the speed of the planet changes as it moves along different segments of its orbit and why.

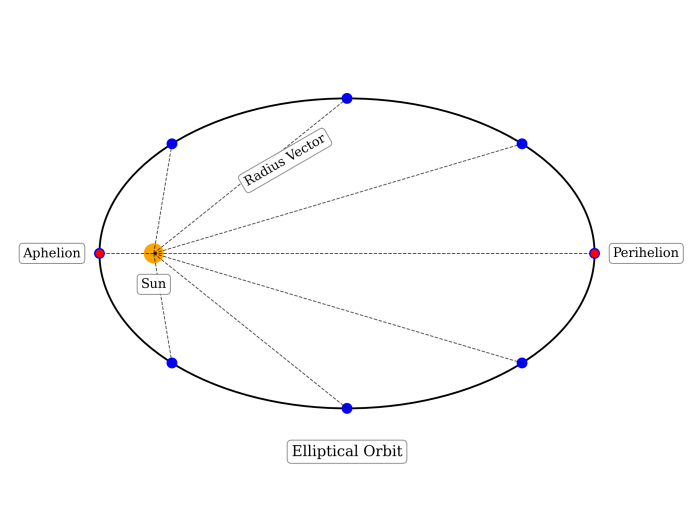

Fig. 1. Diagram for Question 1

Question 2 10.0 points

Refer to the diagram below showing a spacecraft executing a Hohmann transfer orbit between two circular orbits around Earth. Calculate the velocity changes (Δv) required at each segment of the transfer. Provide a detailed step-by-step explanation of the calculation.

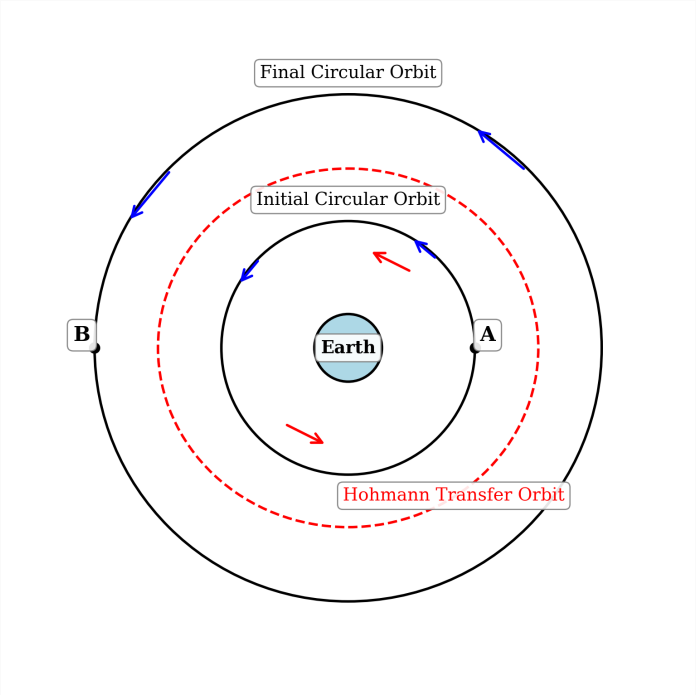

Fig. 2. Diagram for Question 2

Question 3

10.0 points

Analyze the diagram below illustrating the trajectory of a space probe performing a gravitational slingshot maneuver around a planet. Explain how the slingshot alters the probe's trajectory and velocity. Discuss the conservation laws applicable in this maneuver, considering <eq_q_1> and <eq_q_2>.

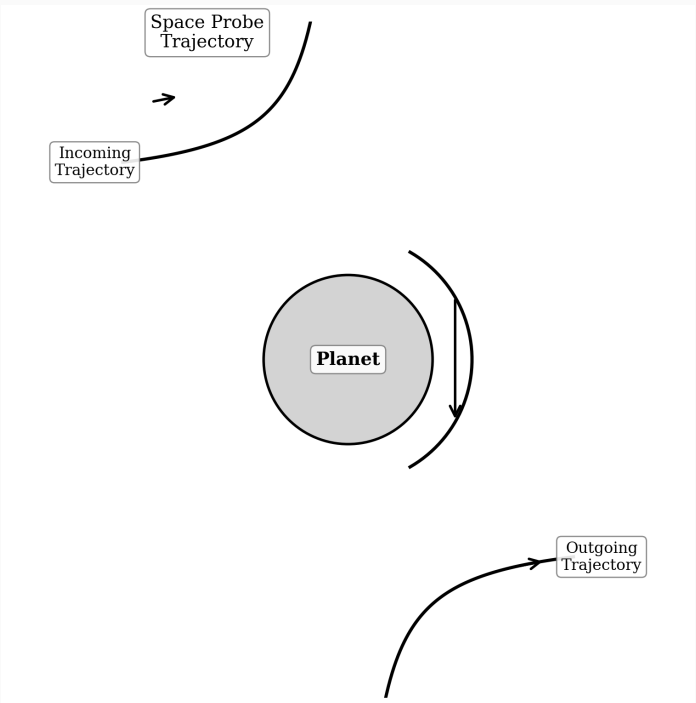

Fig. 3. Diagram for Question 3

Question 4

10.0 points

Examine the diagram below representing the five Lagrange points (L1-L5) in the Earth-Sun system. Describe the stability of each point and how satellites can utilize these positions.

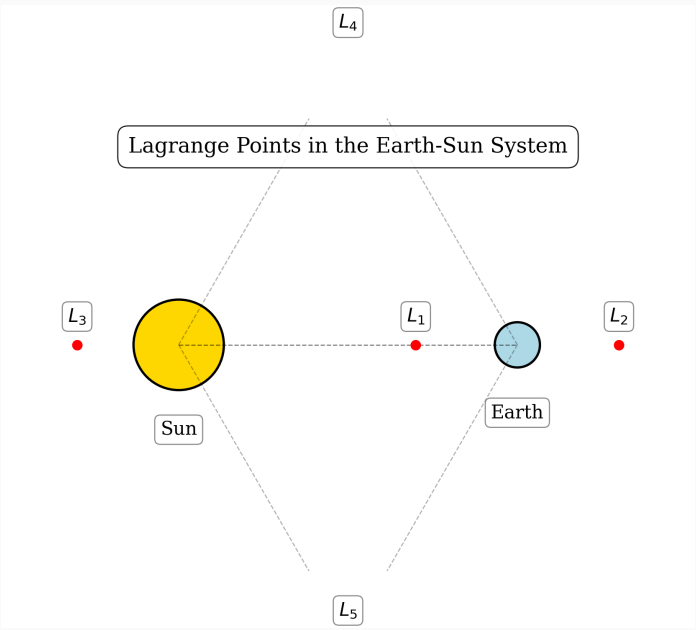

Fig. 4. Diagram for Question 4

Question 5

10.0 points

Based on the diagram below detailing Mercury's orbit, explain the phenomenon of orbital precession and its causes, including both Newtonian mechanics and general relativity contributions, with equations <eq_q_1> and <eq_q_2> resolved.

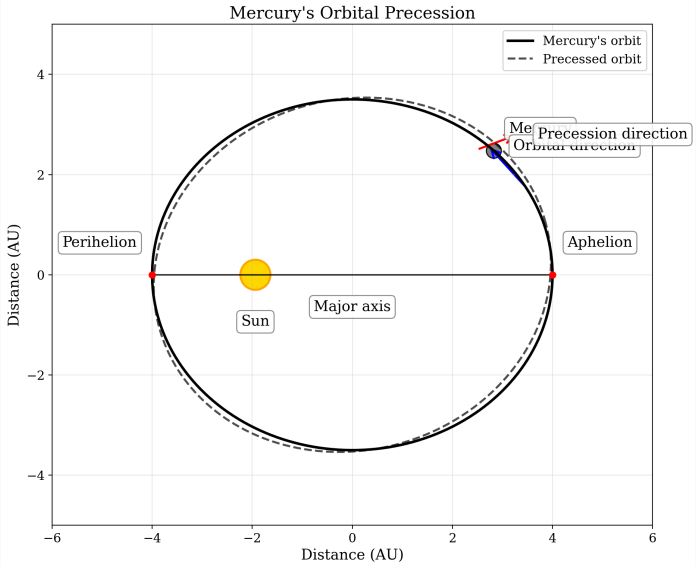

Fig. 5. Diagram for Question 5